

Equation 2 Study Guide

Understanding the paper's original form, your boss's update, and how it maps to Nextnano

Study goal: Goal: understand what Equation 2 is calculating, what every term means, which quantities come from Nextnano versus DFT, and exactly why the k-space and energy-form changes on your boss's slide are useful.

Primary source: Ramesh et al., Enhanced Interband Optical Nonlinearities from Coupled Quantum Wells (2026), Methods Sec. 5.1, pp. 11-12.

Update source: the equation-adjustment slide supplied in this chat.

1. Start with the big picture

One-sentence meaning: Equation 2 converts quantum-well electronic structure into a predicted second-order susceptibility, $\chi^{(2)}$. It combines (1) state energies, (2) wavefunctions and matrix elements, (3) optical resonance, and (4) a sum/integral over in-plane k states.

The calculation chain

Step	What happens	Main output
1	Choose the GaAs/AlGaAs coupled-QW geometry.	Layer widths, compositions, interfaces
2	Solve the quantum problem.	Electron and heavy-hole energies + envelope wavefunctions
3	Build transition quantities.	Overlaps, z matrix elements, transition energies
4	Evaluate the optical denominators.	How close each pathway is to one- and two-photon resonance
5	Integrate over k_{parallel} .	Add the contribution from the in-plane electronic states
6	Apply the prefactor.	Final $\chi^{(2)}$ in physical units

Important distinction before comparing equations

Do not mix stages: Your boss's slide starts from the paper's GENERAL Eq. 1, not from the already-simplified Eq. 2. The slide then makes the k-space integral and the use of energy units explicit. The same ideas can be carried into an Eq. 2 implementation, but the expressions should not be compared term-for-term as if they were already the same stage of the derivation.

- **Paper Eq. 1:** general dipole-matrix expression with sums over polarization, bands, states, and k_{parallel} .
- **Paper Eq. 2:** specialized expression for the coupled-QW xzx susceptibility after Bloch/envelope separation and other assumptions.
- **Boss update:** mainly changes how the k-state sum and frequency denominators are written for numerical work.

2. Paper Equation 2: the form you ultimately want to understand

$$\chi_{xzx}^{(2)}(\omega_1, \omega_2) = \frac{N_z e^3 r_{e, hh}^2}{6 \epsilon_0 \hbar^2} \sum_{k_{\parallel}} \sum_{m, n} \sum_l \left[\frac{A_{mnl}}{D_{mnl}^{(e)}} - \frac{B_{mnl}}{D_{mnl}^{(hh)}} \right]$$

Compact notation used in this guide. A and B are the electron-side and heavy-hole-side numerator pathways; D contains the resonance denominators.

The two numerator pathways

$$A_{mnl} = \langle \psi_{hh, m} | \psi_{e, n} \rangle \langle \psi_{e, n} | Z | \psi_{e, l} \rangle \langle \psi_{e, l} | \psi_{hh, m} \rangle$$

$$B_{mnl} = \langle \psi_{e, n} | \psi_{hh, m} \rangle \langle \psi_{hh, m} | Z | \psi_{hh, l} \rangle \langle \psi_{hh, l} | \psi_{e, n} \rangle$$

The two denominator products

$$D_{mnl}^{(e)} = (\omega_{hh, m}^{e, n}(k_{\parallel}) - \omega_1 - \omega_2 + i\Gamma) (\omega_{hh, m}^{e, l}(k_{\parallel}) - \omega_1 + i\Gamma)$$

$$D_{mnl}^{(hh)} = (\omega_{hh, m}^{e, n}(k_{\parallel}) - \omega_1 - \omega_2 + i\Gamma) (\omega_{hh, l}^{e, n}(k_{\parallel}) - \omega_1 + i\Gamma)$$

Mental model: Read Equation 2 as: prefactor x sum over all relevant states and k points x [electron-side pathway - heavy-hole-side pathway]. Large $\chi^{(2)}$ occurs when the matrix elements are large, the resonance denominators are small, and the two pathways do not cancel.

3. Every term in Equation 2 - part I

Term	Plain-language meaning	Where it comes from / why it matters
$\chi^{(2)}_{\text{xx}}$	Second-order nonlinear susceptibility tensor element.	The final quantity being calculated. For the paper, this is the coupled-QW nonlinear response of interest.
ω_1, ω_2	Input optical angular frequencies.	For SHG, $\omega_1 = \omega_2 = \omega$.
N_z	Number of quantum wells per unit length.	Turns a microscopic QW response into a macroscopic susceptibility density.
e	Elementary charge.	Physical constant in the dipole interaction prefactor.
ϵ_0	Vacuum permittivity.	Electromagnetic normalization constant.
$\hbar$	Reduced Planck constant.	Connects energy and angular frequency through $E = \hbar \omega$.
$r_{e,hh}$	Unit-cell interband matrix element between conduction and heavy-hole Bloch states.	Paper obtains this microscopic quantity from DFT/VASP using HSE06; it is pulled outside the envelope-function sums.
m, n, l	Bound-state indices.	The paper keeps the first two bound states in the electron and heavy-hole bands.

The wavefunctions

- **$\psi_{e,n}(z)$** : electron envelope wavefunction for electron state n .
- **$\psi_{hh,m}(z)$** : heavy-hole envelope wavefunction for heavy-hole state m .
- **Envelope function**: the slowly varying QW-scale part of the semiconductor wavefunction. This is the part determined from the heterostructure quantum calculation.

Why this separation matters: The paper separates the total semiconductor wavefunction into a unit-cell (Bloch) part and an envelope part. The unit-cell interband strength is represented by $r_{e,hh}$; the coupled-QW geometry primarily changes the envelope wavefunctions $\psi(z)$.

4. Every term in Equation 2 - part II

Term	Meaning	Interpretation
$\langle \psi_{hh,m} \psi_{e,n} \rangle$	Electron-heavy-hole envelope overlap.	Large when the electron and hole occupy the same spatial region; small when separated.
$\langle \psi_{e,n} z \psi_{e,l} \rangle$	Electron z-directed position/dipole matrix element.	Measures the electron-side QW dipole along the growth direction.
$\langle \psi_{hh,m} z \psi_{hh,l} \rangle$	Heavy-hole z-directed position/dipole matrix element.	Heavy-hole counterpart of the electron term.
$\omega^{(e,n)(hh,m)}(k_{\text{parallel}})$	Interband transition frequency between hh_m and e_n at a particular in-plane k .	Equivalent transition energy is $E_{e,n}(k) - E_{hh,m}(k) = \hbar \omega_{\text{transition}}$.
Γ	Line broadening; the paper assumes 5 meV.	Gives the resonance a finite width and makes $\chi^{(2)}$ complex rather than singular at exact resonance.
k_{parallel}	In-plane wavevector, $\sqrt{k_x^2 + k_y^2}$.	Carriers are confined in z but can disperse in the QW plane, so transition energies vary with k_{parallel} .

Two matrix elements to be able to recognize instantly

$$\langle \psi_a | \psi_b \rangle = \int \psi_a^*(z) \psi_b(z) dz$$

Overlap integral: asks how much two wavefunctions occupy the same locations in z .

$$\langle \psi_a | z | \psi_b \rangle = \int \psi_a^*(z) z \psi_b(z) dz$$

z matrix element: asks how strongly those states form a dipole along the quantum-well growth direction.

5. What the numerator is doing physically

Electron-side pathway

$$A_{mnl} = \langle \psi_{hh,m} | \psi_{e,n} \rangle \langle \psi_{e,n} | z | \psi_{e,l} \rangle \langle \psi_{e,l} | \psi_{hh,m} \rangle$$

1. Start in a heavy-hole state hh_m .
2. Use an interband overlap to connect hh_m with an electron state e_n .
3. Use the z-directed electron matrix element to connect electron states e_n and e_l .
4. Use another interband overlap to return to the heavy-hole state.

Heavy-hole-side pathway

$$B_{mnl} = \langle \psi_{e,n} | \psi_{hh,m} \rangle \langle \psi_{hh,m} | z | \psi_{hh,l} \rangle \langle \psi_{hh,l} | \psi_{e,n} \rangle$$

5. Start with the electron-heavy-hole interband overlap.
6. Use the z-directed matrix element within the heavy-hole states hh_m and hh_l .
7. Use the second interband overlap to close the pathway.

Cancellation matters: Equation 2 subtracts the heavy-hole-side contribution from the electron-side contribution. Therefore, 'make every matrix element large' is not enough: two large contributions can still cancel.

Why asymmetry is useful

Changing well widths, barrier widths, or interface grading changes $\psi(z)$. That changes both the overlap factors and the z matrix elements. Structural asymmetry can prevent the transition pathways from canceling as strongly as they would in a more symmetric structure.

6. What the denominators are doing: resonance

$$\omega_1 = \omega_2 = \omega, \quad E_1 = E_2 = E_{\text{photon}}, \quad E_1 + E_2 = 2E_{\text{photon}}$$

$$E_{\text{photon}} \approx \frac{1240 \text{ eV nm}}{1550 \text{ nm}} \approx 0.80 \text{ eV}, \quad 2E_{\text{photon}} \approx 1.60 \text{ eV}$$

For 1550-nm SHG, the two-photon / second-harmonic resonance is therefore near 1.60 eV. If a QW interband transition approaches that energy, the first denominator becomes small and the nonlinear response can increase strongly.

Denominator idea	What it asks
Transition - ($\omega_1 + \omega_2$)	Is the final interband transition close to the combined energy of the two input photons?
Transition - ω_1 (or ω_2)	Is the intermediate step close to a one-photon resonance?
+ i Gamma	How broad / damped is that resonance?

Key consequence: This is why a tiny change in layer thickness or interface grading can produce a large change in $\chi^{(2)}$: the structure shifts the state energies and can move a denominator toward or away from resonance.

7. $k_{\parallel}$: why a single $k = 0$ calculation is not the whole answer

The QW confines the carriers in z , but not in the x - y plane. Therefore the electron and heavy-hole energies disperse with in-plane wavevector $k_{\parallel}$. Each k state can have a different transition energy and thus a different resonance contribution.

$$\sum_{k_{\parallel}} F(k_{\parallel}) \rightarrow \iint F(k_x, k_y) dk_x dk_y \rightarrow 2\pi \int_0^{k_{\max}} F(k_{\parallel}) k_{\parallel} dk_{\parallel}$$

Why the factor $k_{\parallel}$ appears

In the k_x - k_y plane, radial integration is a sum over concentric rings. A ring at larger k has a larger circumference, so it represents more states. In polar coordinates:

$$dk_x dk_y = k_{\parallel} dk_{\parallel} d\theta, \text{ and } \int_0^{2\pi} d\theta = 2\pi.$$

Numerical takeaway: Practical implication: equal spacing in k does NOT mean equal weight. A radial quadrature typically needs a weight proportional to $k_{\parallel} \cdot \Delta k$ (plus whatever normalization convention the derivation requires).

The paper states that the k -state contribution saturated by about one-tenth of the Brillouin zone, so the authors integrated from $k_{\parallel} = 0$ to that cutoff.

8. Your boss's slide: what is being changed

Adjusting the form

$$\chi_{ijk}^{(2)}(\omega_1, \omega_2) = \frac{N_z e^3}{2\epsilon_0 \hbar^2} \sum_P \sum_{b_1, b_2, b_3} \sum_{l, m, n} f_{b,l} \sum_{k_{||}} \frac{\langle \phi_{1,l}^* | r_i | \phi_{2,m} \rangle \langle \phi_{2,m}^* | r_j | \phi_{3,n} \rangle \langle \phi_{3,n}^* | r_k | \phi_{1,l} \rangle}{[\omega_{b_2,m}(k_{||}) - \omega_{b_1,l}(k_{||}) - \omega_1 - \omega_2][\omega_{b_3,n}(k_{||}) - \omega_{b_1,l}(k_{||}) - \omega_2]}$$

$$\chi_{ijk}^{(2)} = \frac{N_z e^3}{2\epsilon_0 \hbar} \sum_P \sum_{b_1, b_2, b_3} \sum_{l, m, n} f_{b,l} \iint \frac{\langle \phi_{1,l}^* | r_i | \phi_{2,m} \rangle \langle \phi_{2,m}^* | r_j | \phi_{3,n} \rangle \langle \phi_{3,n}^* | r_k | \phi_{1,l} \rangle}{[\hbar\omega_{b_2,m}(k_{||}) - \hbar\omega_{b_1,l}(k_{||}) - \hbar\omega_1 - \hbar\omega_2][\hbar\omega_{b_3,n}(k_{||}) - \hbar\omega_{b_1,l}(k_{||}) - \hbar\omega_2]} dk_x dk_y$$

$$\chi_{ijk}^{(2)} = \frac{N_z e^3}{2\epsilon_0 \hbar} \sum_P \sum_{b_1, b_2, b_3} \sum_{l, m, n} f_{b,l} \iint \frac{\langle \phi_{1,l}^* | r_i | \phi_{2,m} \rangle \langle \phi_{2,m}^* | r_j | \phi_{3,n} \rangle \langle \phi_{3,n}^* | r_k | \phi_{1,l} \rangle}{[E_{b_2,m}(k_{||}) - E_{b_1,l}(k_{||}) - E_1 - E_2][E_{b_3,n}(k_{||}) - E_{b_1,l}(k_{||}) - E_2]} dk_x dk_y$$

$$\chi_{ijk}^{(2)} = \frac{N_z e^3}{2\epsilon_0 \hbar} \sum_P \sum_{b_1, b_2, b_3} \sum_{l, m, n} f_{b,l} \int \frac{\langle \phi_{1,l}^* | r_i | \phi_{2,m} \rangle \langle \phi_{2,m}^* | r_j | \phi_{3,n} \rangle \langle \phi_{3,n}^* | r_k | \phi_{1,l} \rangle}{[E_{b_2,m}(k_{||}) - E_{b_1,l}(k_{||}) - E_1 - E_2][E_{b_3,n}(k_{||}) - E_{b_1,l}(k_{||}) - E_2]} 2\pi k_{||} dk_{||}$$

Equation-adjustment slide supplied by the user.

Main idea: Most of the slide is a change of representation, not a new physical model. The matrix-element numerator, band/state sums, and occupation factor are retained. The main changes are: make the k sum an explicit integral, express detunings in energies, and reduce a 2D integral to a 1D radial integral under an isotropy assumption.

Slide change	Before	After	Why useful
k states	sum over k_parallel	double integral dk_x dk_y	Makes continuum weighting explicit.
Optical variables	angular frequencies omega	energies E = hbar omega	Matches simulation outputs that are naturally handled in eV.
2D integral	dk_x dk_y	2pi k_parallel dk_parallel	Reduces 2D integration to 1D if the problem is rotationally symmetric in-plane.

9. Boss update - Step 1: replace the k sum with a continuum integral

$$\sum_{k_{\parallel}} F(k_{\parallel}) \rightarrow \iint F(k_x, k_y) dk_x dk_y \rightarrow 2\pi \int_0^{k_{\max}} F(k_{\parallel}) k_{\parallel} dk_{\parallel}$$

What changed?

Paper notation: sum over k_{parallel} . **Boss notation:** an explicit integral over the in-plane coordinates k_x and k_y .

Why make this change?

- It tells you exactly what numerical object must be integrated.
- It exposes the k-space density/weighting rather than hiding it inside a symbolic sum.
- It makes it easier to connect a discrete Nextnano k grid to a numerical quadrature.

How this maps to a discrete Nextnano grid

For a radial grid $k_0, k_1, \dots, k_N$, a numerical implementation is conceptually:

```
for each  $k_i$ :
    obtain transition energies and required matrix elements
    evaluate the Eq. 2 integrand  $F(k_i)$ 
    multiply by the k-space quadrature weight
sum all weighted contributions
```

Why it may matter numerically: This step can change the absolute magnitude if the original code gave every k point equal weight or used a different continuum normalization.

10. Boss update - Step 2: change frequency detunings into energy detunings

$$\hbar\omega = E, \quad \hbar(\omega_{tr} - \omega_1 - \omega_2) = E_{tr} - E_1 - E_2$$

What changed?

A denominator written as a frequency mismatch, such as $\omega_{\text{transition}} - \omega_1 - \omega_2$, is rewritten using $E = \hbar\omega$ as an energy mismatch:

$$E_{\text{transition}} - E_1 - E_2$$

Why this is easier with Nextnano

- Nextnano state energies are naturally reported as energies (commonly eV).
- Your optical photon energy can also be written in eV.
- The denominator can then be assembled from quantities in one consistent unit system instead of repeatedly converting $E \leftrightarrow \omega$.

Direct simulation connection: Example: if Nextnano gives E_{e1} and E_{hh1} , the optical transition energy is $\Delta E_{11} = E_{e1} - E_{hh1}$. That ΔE is what you compare with $2E_{\text{photon}}$ for the SHG resonance.

What is NOT supposed to change

If the conversion is done consistently, changing from frequency units to energy units should not change the physics. It is a unit/representation change. A different numerical result would signal that a prefactor, $\hbar$ factor, or unit conversion was also changed.

11. Boss update - Step 3: reduce the 2D integral to a radial integral

$$\sum_{k_{\parallel}} F(k_{\parallel}) \rightarrow \iint F(k_x, k_y) dk_x dk_y \rightarrow 2\pi \int_0^{k_{\max}} F(k_{\parallel}) k_{\parallel} dk_{\parallel}$$

The assumption

This reduction is valid when the integrand depends only on the magnitude $k_{\parallel} = \sqrt{k_x^2 + k_y^2}$, not on the in-plane angle θ .

The geometry

- Start with $dk_x dk_y$.
- Change to polar coordinates: $dk_x dk_y = k_{\parallel} dk_{\parallel} d\theta$.
- If there is no θ dependence, integrate θ from 0 to 2π .
- That leaves $2\pi k_{\parallel} dk_{\parallel}$.

Why your boss likely wants it: This is computationally valuable because it can turn a 2D k_x - k_y calculation into a 1D radial integration. You still represent the full ring of states through the $2\pi k$ factor.

What can go wrong

If the band structure or matrix elements are anisotropic in-plane, the θ dependence cannot simply be integrated out as 2π . Then a true 2D treatment may be needed.

12. How the updated form helps your Nextnano workflow

Quantity needed by $\chi^{(2)}$	Natural source	What you do with it
$E_{e,n}(k_{\text{parallel}})$	Nextnano	Electron energies versus k.
$E_{hh,m}(k_{\text{parallel}})$	Nextnano	Heavy-hole energies versus k.
$\psi_{e,n}(z), \psi_{hh,m}(z)$	Nextnano	Build overlaps and z matrix elements.
$\Delta E_{e,hh}(k)$	Post-processing	Subtract heavy-hole energy from electron energy at the same k.
$\langle hh e \rangle$ and $\langle e z e' \rangle / \langle hh z hh' \rangle$	Post-processing from wavefunctions	Construct the Eq. 2 numerator.
$r_{e,hh}$	DFT/VASP HSE06 in the paper	Atomic/unit-cell interband strength.
E_1, E_2	Laser wavelength	Photon energies for the denominator.
Gamma	Model parameter; paper uses 5 meV	Broadening in the complex denominators.
k-space weight	Chosen integration convention	Converts the k grid into the continuum integral.

A practical mental workflow

8. Generate the QW electronic structure in Nextnano.
9. For each k_{parallel} , identify the $e1/e2$ and $hh1/hh2$ states consistently.
10. Compute the interband transition energies $\Delta E_{nm}(k)$.
11. Compute the overlap and z matrix elements from the envelope wavefunctions.
12. Form the electron-side and heavy-hole-side Eq. 2 contributions.
13. Evaluate the complex resonance denominators using photon energies and Gamma.
14. Apply the radial k-space weight.
15. Integrate from $k = 0$ to the chosen k_{max} and check convergence.
16. Apply the remaining prefactors using one fully consistent unit system.

Why it is practical: The update is useful because Steps 3, 6, and 7 now line up naturally with what a k-resolved Nextnano + Python workflow actually stores: energies in eV, values on a k grid, and a numerical integral.

13. Two algebra/normalization items to verify before coding the update

Important: These are not claims that the slide is wrong. They are places where the displayed slide alone does not show enough detail to reconstruct the complete normalization, so they should be verified before changing production code.

Check A - the k-space normalization

$$\frac{1}{(2\pi)^2} \iint d^2k = \frac{1}{2\pi} \int_0^{k_{\max}} k_{\parallel} dk_{\parallel}$$

A continuum state sum is often normalized with a factor such as $1/(2\pi)^2$ per unit area. If that convention applies, the radial measure is not simply $2\pi k dk$; the normalization changes the net factor. The paper's printed Eq. 2 does not explicitly display this continuum measure, so the exact convention must be traced through the derivation/code.

Magnitude sensitivity: Why it matters: 2π and $1/(2\pi)$ differ by $(2\pi)^2$ approximately 39.48. A missing or duplicated normalization can therefore produce a very large error in the absolute χ^2 .

Check B - the remaining $\hbar$ in the slide's energy form

$$\frac{1}{\hbar^2} \frac{1}{\Delta\omega_1 \Delta\omega_2} = \frac{1}{\Delta E_1 \Delta E_2} \quad \text{if} \quad \Delta E = \hbar \Delta\omega$$

Based only on the displayed algebra, converting BOTH frequency denominators to energies supplies two factors of $\hbar$, which cancel the original $1/\hbar^2$ prefactor. The lower lines on the slide visibly retain a $1/\hbar$ prefactor. That may reflect another convention or normalization step not shown on the slide, but the source of that remaining factor should be documented before implementation.

Ask your boss: Best question to ask: 'When both detunings are converted with $\Delta E = \hbar \Delta\omega$, what additional step produces the remaining $1/\hbar$ in the prefactor?'

14. Do not accidentally lose Eq. 2-specific physics when applying the update

Because the boss slide begins from the general Eq. 1, several Eq. 2-specific simplifications are not visible on that slide. When you transfer the k-space and energy changes to your Eq. 2 code, keep track of the following:

Eq. 2 feature	Why it is important
r_{e,hh^2} is pulled outside the envelope-function sums	This comes from the Bloch/envelope separation and represents the unit-cell interband strength.
First two bound states are retained	The paper's Eq. 2 is specialized to $e1/e2$ and $hh1/hh2$.
Filled valence-band occupation is taken as unity	The $f_{b,l}$ factor from Eq. 1 is no longer an active variable in Eq. 2.
Electron-side minus heavy-hole-side pathways	The subtraction and cancellation behavior are central to the specialized expression.
+ i Gamma broadening	The paper's Eq. 2 includes finite line broadening (5 meV). The boss's displayed general form does not show this term.

Implementation strategy: The safe approach is: preserve the specialized Eq. 2 numerator and broadening physics, then change only the k integration and the frequency-versus-energy representation after you have verified the corresponding prefactors.

15. Final review sheet - what to remember

Core equation: Equation 2 = prefactor x k-space integral x [electron-side nonlinear pathway - heavy-hole-side nonlinear pathway].

The four knobs that most directly control $\chi^{(2)}$

Knob	What changes it	Effect
Transition energies	Layer widths, composition, interfaces	Moves resonance denominators closer to or farther from zero.
Electron-hole overlaps	Wavefunction localization	Controls interband envelope coupling.
z matrix elements	Asymmetry and state localization	Controls the QW dipole along the growth direction.
k-space weighting / dispersion	Band structure + numerical integration	Determines how contributions away from $k = 0$ accumulate.

Your boss's three main changes

17. sum over k_{parallel} -> explicit integral over k_x and k_y .
18. frequency detunings -> energy detunings using $E = \hbar \omega$.
19. 2D Cartesian integral -> 1D radial integral with $2\pi k_{\text{parallel}} dk_{\text{parallel}}$ when in-plane isotropy is assumed.

Why those changes help

- They make the numerical integration explicit.
- They map more naturally to Nextnano energies in eV.
- They reveal the required k-dependent quadrature weighting.
- They can reduce a 2D integral to a much cheaper 1D integral.

Two checks before trusting an absolute pm/V number

- Confirm the exact k-space normalization, including any $(2\pi)^2$ factor.
- Confirm the $\hbar$ prefactor after converting both denominator factors from angular frequency to energy.

Mastery test: If you can explain the numerator, the two resonance denominators, the k_{parallel} weight, and where each input comes from, you understand the operational content of Equation 2 well enough to audit a Nextnano implementation.

Source note

Paper basis: Ramesh et al., Enhanced Interband Optical Nonlinearities from Coupled Quantum Wells (2026), especially Methods Sec. 5.1 (paper Eq. 1 and Eq. 2, pp. 11-12) and Sec. 3.2 for the stated physical role of diagonal intersubband matrix elements.

Update basis: the equation-adjustment slide supplied by the user. Interpretive comments about the unexplained residual $\hbar$ and possible k-space normalization are explicitly labeled as checks rather than as statements from the paper.